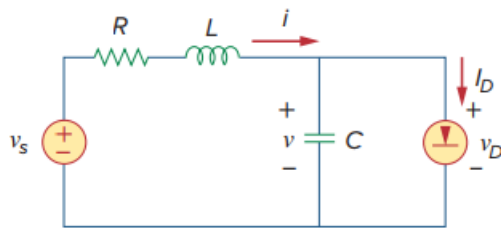


Problem

Figure 8.124 shows a typical tunnel-diode oscillator circuit. The diode is modeled as a nonlinear resistor with $i_D = f(v_D)$, i.e., the diode current is a nonlinear function of the voltage across the diode. Derive the differential equation for the circuit in terms of v and i_D .

Figure 8.124:



Step-by-step solution

Step 1 of 3

Refer to Figure 8.124 in the textbook.

Draw the modified circuit with current and voltage notations.

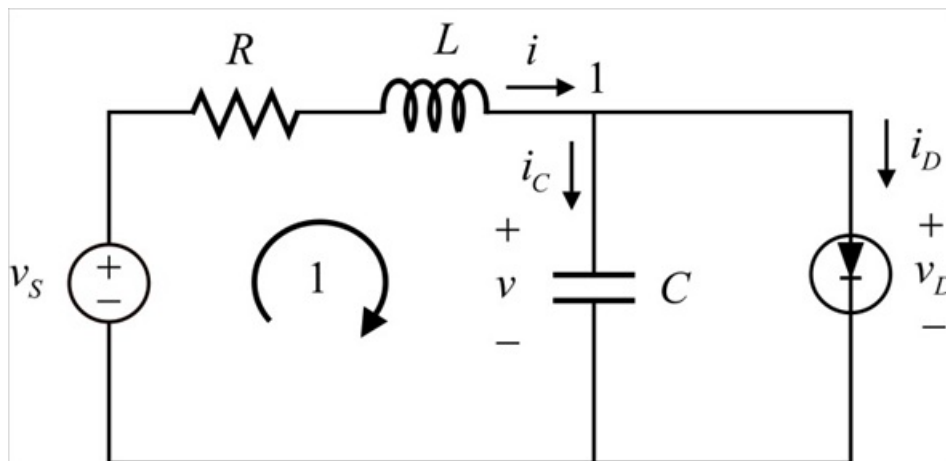


Figure 1

Step 2 of 3

Consider the expression for the current through the capacitor.

$$i_c = C \frac{dv}{dt}$$

Consider the expression for voltage across the inductor.

$$v_L = L \frac{di}{dt}$$

Consider the expression for voltage across the resistor.

$$v_R = iR$$

Write Kirchhoff's current law at the node 1.

$$i = i_c + i_D$$

Substitute $C \frac{dv}{dt}$ for i_c .

$$i = C \frac{dv}{dt} + i_D$$

Step 3 of 3

Apply Kirchhoff's voltage law for the Loop 1.

$$v_R + v_L + v = v_s$$

Substitute iR for v_R , $L \frac{di}{dt}$ for v_L .

$$iR + L \frac{di}{dt} + v = v_s$$

Substitute $C \frac{dv}{dt} + i_D$ for i .

$$\left[C \frac{dv}{dt} + i_D \right] R + L \frac{d}{dt} \left[C \frac{dv}{dt} + i_D \right] + v = v_s$$

$$RC \frac{dv}{dt} + Ri_D + LC \frac{d^2v}{dt^2} + L \frac{di_D}{dt} + v = v_s$$

$$LC \frac{d^2v}{dt^2} + RC \frac{dv}{dt} + Ri_D + L \frac{di_D}{dt} = v_s - v$$

Divide with LC .

$$\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{R}{LC} i_D + \frac{1}{C} \frac{di_D}{dt} = \frac{v_s - v}{LC}$$

Therefore, the differential equation for the circuit in terms of v and i_D is,

$$\boxed{\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{R}{LC} i_D + \frac{1}{C} \frac{di_D}{dt} = \frac{v_s - v}{LC}}$$